TYPICAL UNIT 4 CALCULATION MCQ'S IN ME 265 - DETAILED SOLUTIONS

1. $T = T_{sat@350\ kPa} = 138.86\ ℃ \qquad v = v_{g@350\ kPa} = 0.52422\ m^3/kg$

$$m = \frac{V}{v} = \frac{0.6\ m^3}{0.52422\ m^3/kg} = \mathbf{0.114\ kg}$$

2. $P = P_{sat@90\ ℃} = 70.183\ kPa$

But at 90 ℃, $v_f = 0.001036\ m^3/kg$ and $v_g = 2.3593\ m^3/kg$

Also, $V = V_f + V_g = m_f v_f + m_g v_g = 8 \times 0.001036 + 2 \times 2.3593 = \mathbf{4.73\ m^3}$

3. $v = \frac{V}{m} = \frac{0.08}{4} = 0.02\ m^3/kg$

But at 160 kPa, $v_f = 0.0007437\ m^3/kg$ and $v_g = 0.12348\ m^3/kg$

But, $v_f < v < v_g$ hence, $x = \dfrac{v - v_f}{v_{fg}} = \dfrac{0.02 - 0.0007437}{0.12348 - 0.0007437} = \mathbf{0.157}$

4. $v = {RT}/{P} = \dfrac{0.0815\frac{kJ}{kgK} \times 323.15}{1000\ kPa} = \mathbf{0.026369}$

5. $P = P_{sat@90\ ℃} = 70.183\ kPa \qquad v = v_{f@90℃} = 0.0010369\ m^3/kg$

Total volume of tank $= mv_f = 50\ kg \times 0.001036\ m^3/kg = \mathbf{0.0518\ m^3}$

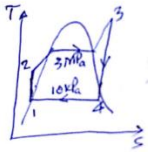
6. $P = P_{sat@90\ ℃} = 70.183\ kPa$

7. $v = \frac{V}{m} = \frac{0.08}{4} = 0.02\ m^3/kg$

but at 160 kPa, $v_f = 0.0007437\ m^3/kg$ and $v_g = 0.12348\ m^3/kg$

since $v_f < v < v_g$, $T = T_{sat@160\ kPa} = \mathbf{-15.60}\ ℃$

8. $P_3 = 3\ MPa, \quad T_3 = 600\ ℃, \quad h_3 = 3682.8\frac{kJ}{kg}, \quad s_3 = s_4 = 7.5103$



$P_4 = 10\ kPa, \quad s_g = 8.1488 < s_4$ (saturated liquid vapour mixture)

$s_f = 0.6492\ kJ/kgK \qquad s_{fg} = 7.4996\ kJ/kgK$

$\therefore x_4 = \frac{7.5103 - 0.6494}{7.4996} = 0.915; \quad h_4 = 191.81 + 0.915(2392.1) = 2380.6\ kJ/kg\ K$

$q_{in} = h_3 - h_1 = 3491\ kJ/kgK \quad q_{out} = h_4 - h_1 = 2380.6 - 191.81 = 2188.8\ kJ/kgK$

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2188.8}{3491} = 0.3730 = \mathbf{37.3}\ \%$$

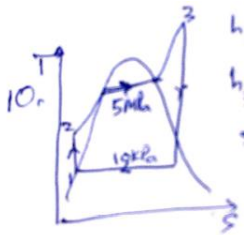
9. $P_3 = 5\ MPa, 600\ ℃ \qquad s_3 = 7.2605\ kJ/kgK \qquad x_4 = \frac{7.2605 - 0.6492}{7.4996} = 0.882$

$\therefore 1 - x_4 = 1 - 0.882 = 0.118 = 11.8\ \% \cong \mathbf{12}\ \%$

10. $h_2 = h_1 = h_{f@10\ kPa} = 191.81\ kJ/kgK$

$h_3 = h_{@5MPa,600℃} = 3666.9\ kJ/kgK$

$s_3 = s_{@5MPa,600℃} = 7.2605\ kJ/kgK$



$$x_4 = \frac{7.2605 - 0.6492}{7.4996} = 0.882$$

$h_4 = 191.81 + 0.882(2392.1) = 2301.8\ kJ/kg\ K$

$$\dot{m} = \frac{300\ kg/s}{(3666.9 - 191.81)} = 0.0863\ kg/s$$

$\left|\dot{W}\right| = \dot{m}(h_3 - h_4) = 0.0863(3666.9 - 2301.8) \cong \mathbf{118\ kW}$